The Dot Product

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Computing dot products and angles**

Let $\mathbf{u} = \langle 2, 5\rangle$ and $\mathbf{v} = \langle -3, 4\rangle$. Find:

a $\mathbf{u} \cdot \mathbf{v}$ $(2)(-3) + (5)(4) = 14$.

b $|\mathbf{u}|$ and $|\mathbf{v}|$ $|\mathbf{u}| = \sqrt{29}$, $|\mathbf{v}| = 5$.

c the angle between $\mathbf{u}$ and $\mathbf{v}$, to one decimal place $\cos\theta = \frac{14}{5\sqrt{29}} \approx 0.5199$, so $\theta \approx 58.7°$.

(a) $(2)(-3) + (5)(4) = 14$.; (b) $|\mathbf{u}| = \sqrt{29}$, $|\mathbf{v}| = 5$.; (c) $\cos\theta = \frac{14}{5\sqrt{29}} \approx 0.5199$, so $\theta \approx 58.7°$.

2 **[Foundational]** • **Target: Computing dot products and angles**

Determine whether each pair of vectors is orthogonal:

a $\langle 4, -6\rangle$ and $\langle 3, 2\rangle$ $12 - 12 = 0$: orthogonal. **b** $\langle 1, 2\rangle$ and $\langle 2, 1\rangle$ $2 + 2 = 4 \neq 0$: not orthogonal.

(a) $12 - 12 = 0$: orthogonal.; (b) $2 + 2 = 4 \neq 0$: not orthogonal.

3 **[Foundational]** • **Target: Computing dot products and angles**

Find the value of k so that $\langle k, 3\rangle$ is orthogonal to $\langle 2, -4\rangle$.

$2k - 12 = 0$, so $k = 6$.

4 **[Foundational]** • **Target: Projections and work**

Find the vector projection of $\mathbf{u} = \langle 6, 2\rangle$ onto $\mathbf{v} = \langle 3, 0\rangle$.

$\text{proj}_{\mathbf{v}}\mathbf{u} = \frac{\mathbf{u}\cdot\mathbf{v}}{|\mathbf{v}|^2}\mathbf{v} = \frac{18}{9}\langle 3, 0\rangle = \langle 6, 0\rangle$.

5 **[Foundational]** • **Target: Projections and work**

A force $\mathbf{F} = \langle 8, 5\rangle$ N moves an object along the displacement $\mathbf{d} = \langle 10, 0\rangle$ m. Find the work done.

$W = \mathbf{F} \cdot \mathbf{d} = 80$ J.

6 **[Foundational]** • **Target: Projections and work**

A wagon is pulled a distance of 20 m by a rope making an angle of $30°$ with the ground, with a force of magnitude 50 N. Find the work done, to one decimal place.

$W = |\mathbf{F}||\mathbf{d}|\cos\theta = 50(20)\cos 30° \approx 1000(0.8660) \approx 866.0$ J.

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7 [Foundational] • Target: Properties of the dot product

Let $\mathbf{u} = \langle 3, 1\rangle$, $\mathbf{v} = \langle -2, 4\rangle$, $\mathbf{w} = \langle 1, 1\rangle$. Verify that $\mathbf{u} \cdot (\mathbf{v} + \mathbf{w}) = \mathbf{u} \cdot \mathbf{v} + \mathbf{u} \cdot \mathbf{w}$ (the distributive property).

$\mathbf{v} + \mathbf{w} = \langle -1, 5\rangle$: $\mathbf{u} \cdot (\mathbf{v} + \mathbf{w}) = -3 + 5 = 2$. Separately: $\mathbf{u} \cdot \mathbf{v} = -6 + 4 = -2$, $\mathbf{u} \cdot \mathbf{w} = 3 + 1 = 4$; sum $= 2$. Both equal 2. ✓

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Properties of the dot product

Show that $\mathbf{u} \cdot \mathbf{u} = |\mathbf{u}|^2$ for $\mathbf{u} = \langle 4, -3\rangle$.

$\mathbf{u} \cdot \mathbf{u} = 16 + 9 = 25$. $|\mathbf{u}|^2 = (\sqrt{25})^2 = 25$. Equal, as expected.

9 [Intermediate] • Target: Angle classification

Without finding the angle exactly, classify each pair of vectors as forming an acute, right, or obtuse angle, using only the sign of the dot product:

a $\langle 3, 5\rangle$ and $\langle 2, -1\rangle$ Dot product $= 6 - 5 = 1 > 0$: acute.

b $\langle -1, 4\rangle$ and $\langle 8, 2\rangle$ Dot product $= -8 + 8 = 0$: right angle.

c $\langle 2, 2\rangle$ and $\langle -3, -3\rangle$ Dot product $= -6 - 6 = -12 < 0$: obtuse.

(a) Dot product $= 6 - 5 = 1 > 0$: acute.; (b) Dot product $= -8 + 8 = 0$: right angle.; (c) Dot product $= -6 - 6 = -12 < 0$: obtuse.

10 [Intermediate] • Target: Scalar projection and further applications

Find the scalar projection of $\mathbf{u} = \langle 5, -2\rangle$ onto $\mathbf{v} = \langle 3, 4\rangle$ (the signed length $\text{comp}_{\mathbf{v}}\mathbf{u} = \frac{\mathbf{u} \cdot \mathbf{v}}{|\mathbf{v}|}$), to two decimal places.

$\mathbf{u} \cdot \mathbf{v} = 15 - 8 = 7$. $|\mathbf{v}| = 5$. $\text{comp}_{\mathbf{v}}\mathbf{u} = \frac{7}{5} = 1.40$.

11 [Intermediate] • Target: Scalar projection and further applications

A sled is pulled by a rope at $40°$ above horizontal with a force of 120 N, moving the sled 15 m horizontally. Find the work done by the horizontal component of the force.

$W = |\mathbf{F}|\cos(40°) \cdot d = 120\cos 40° \cdot 15 \approx 120(0.766)(15) \approx 1378.8$ J.

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12 [Intermediate] • Target: Decomposing a vector into parallel and perpendicular parts

Decompose $\mathbf{u} = \langle 7, 4 \rangle$ into a component parallel to $\mathbf{v} = \langle 1, 0 \rangle$ and a component perpendicular to $\mathbf{v}$.

Parallel: $\text{proj}_{\mathbf{v}}\mathbf{u} = \frac{7}{1}\langle 1, 0 \rangle = \langle 7, 0 \rangle$. Perpendicular: $\mathbf{u} - \langle 7, 0 \rangle = \langle 0, 4 \rangle$. Check: $\langle 7, 0 \rangle + \langle 0, 4 \rangle = \langle 7, 4 \rangle$ ✓, and $\langle 0, 4 \rangle \cdot \langle 1, 0 \rangle = 0$ ✓.

13 [Intermediate] • Target: Deriving the dot product angle formula

State the formula relating the dot product to the angle between two vectors, and explain why $\cos\theta = 0$ corresponds to orthogonal vectors.

$\mathbf{u} \cdot \mathbf{v} = |\mathbf{u}||\mathbf{v}|\cos\theta$. Since $|\mathbf{u}|, |\mathbf{v}| > 0$ for nonzero vectors, the dot product is zero exactly when $\cos\theta = 0$, i.e. $\theta = 90°$ — perpendicular vectors.

14 [Intermediate] • Target: The Cauchy-Schwarz inequality (preview)

Explain why $|\mathbf{u} \cdot \mathbf{v}| \leq |\mathbf{u}||\mathbf{v}|$ must always hold, using the formula $\mathbf{u} \cdot \mathbf{v} = |\mathbf{u}||\mathbf{v}|\cos\theta$.

Since $-1 \leq \cos\theta \leq 1$ for any angle, multiplying by the non-negative quantity $|\mathbf{u}||\mathbf{v}|$ gives $-|\mathbf{u}||\mathbf{v}| \leq \mathbf{u} \cdot \mathbf{v} \leq |\mathbf{u}||\mathbf{v}|$, which is exactly $|\mathbf{u} \cdot \mathbf{v}| \leq |\mathbf{u}||\mathbf{v}|$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Dot products with variable components • Cross-Lesson: 1.02 Domain and Range

Find all values of k such that $\langle k, 3 \rangle$ and $\langle k, -3 \rangle$ are orthogonal.

Dot product: $k^2 - 9 = 0 \Rightarrow k = \pm 3$.

16 [Challenging] • Target: Dot products with variable components • Cross-Lesson: 1.03 Average Rate of Change

Find all values of k such that the angle between $\langle 1, k \rangle$ and $\langle 2, 1 \rangle$ is $45°$.

$\cos 45° = \frac{2+k}{\sqrt{1+k^2}\sqrt{5}} = \frac{\sqrt{2}}{2}$. Squaring: $\frac{(2+k)^2}{5(1+k^2)} = \frac{1}{2}$, so $2(2+k)^2 = 5(1+k^2)$: $2(4+4k+k^2) = 5+5k^2$, $8+8k+2k^2 = 5+5k^2$, $3k^2 - 8k - 3 = 0$. Quadratic formula: $k = \frac{8 \pm \sqrt{64+36}}{6} = \frac{8 \pm 10}{6}$: $k = 3$ or $k = -\frac{1}{3}$.

17 [Challenging] • Target: A capstone dot product application • Cross-Lesson: 2.01 Right-Triangle Trigonometry

A weather vane experiences wind force $\mathbf{F} = \langle 15, -8 \rangle$ N. The vane can only rotate about a pivot, so only the component of force perpendicular to the vane's current direction $\mathbf{d} = \langle 3, 4 \rangle$ causes it to turn. Find the magnitude of the perpendicular (turning) component of the force, using the fact that the perpendicular component's magnitude equals $\frac{|\mathbf{F} \times \mathbf{d}|}{|\mathbf{d}|}$ in 2D (using the scalar cross product $F_x d_y - F_y d_x$), and separately confirm the parallel component using the dot product.

Scalar cross product: $F_x d_y - F_y d_x = 15(4) - (-8)(3) = 60 + 24 = 84$. Perpendicular component magnitude $= \frac{84}{5} = 16.8$ N. Parallel component via dot product: $\mathbf{F} \cdot \mathbf{d} = 45 - 32 = 13$; magnitude $= \frac{13}{5} = 2.6$ N. Together these decompose the total force into turning and non-turning parts.

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18 [Challenging] • Target: A capstone dot product application • Cross-Lesson: 2.02 Radian Measure and Arc Length

Two salespeople's monthly sales across three product categories are given by vectors $\mathbf{s}_1 = \langle 40, 25, 10 \rangle$ and $\mathbf{s}_2 = \langle 15, 30, 45 \rangle$ (units sold). Find $\mathbf{s}_1 \cdot \mathbf{s}_2$ and explain what this dot product does NOT directly tell you about their relative performance (versus what the individual sums would show).

$\mathbf{s}_1 \cdot \mathbf{s}_2 = 40(15) + 25(30) + 10(45) = 600 + 750 + 450 = 1800$. This dot product is a single combined measure of alignment/overlap in their sales patterns, but it does NOT directly reveal who sold more in total (that would come from summing each vector's own components: 75 vs 90 units) — the dot product measures a different kind of relationship (similarity in distribution across categories), not total volume.

19 [Challenging] • Target: Scalar projection and further applications • Cross-Lesson: 2.03 Graphs of Sine and Cosine Functions

Decompose the vector $\mathbf{u} = \langle 6, 2 \rangle$ into components parallel and orthogonal to $\mathbf{v} = \langle 3, 4 \rangle$.

Step 1: Compute dot product and squared magnitude: $\mathbf{u} \cdot \mathbf{v} = (6)(3) + (2)(4) = 18 + 8 = 26$. $||\mathbf{v}||^2 = 3^2 + 4^2 = 9 + 16 = 25$. Step 2: Vector projection (parallel component): $\mathbf{u}_{\parallel} = \text{proj}_{\mathbf{v}}\mathbf{u} = \left(\frac{\mathbf{u} \cdot \mathbf{v}}{||\mathbf{v}||^2}\right)\mathbf{v} = \frac{26}{25}\langle 3, 4 \rangle = \left\langle \frac{78}{25}, \frac{104}{25} \right\rangle = \langle 3.12, 4.16 \rangle$. Step 3: Orthogonal component.

20 [Challenging] • Target: Decomposing a vector into parallel and perpendicular parts • Cross-Lesson: 2.04 Law of Sines and Law of Cosines

Consider two vectors, $\mathbf{u} = (3; k)$ and $\mathbf{v} = (-2; 4)$. If $\mathbf{u}$ is parallel to $\mathbf{v}$, find the value of the scalar parameter k.

Two vectors are parallel if one is a scalar multiple of the other. Therefore, there must exist a scalar c such that $\mathbf{u} = c\mathbf{v}$. This gives us the equations: $3 = c(-2)$ $k = c(4)$ From the first equation, we can solve for c: $c = \frac{3}{-2} = -\frac{3}{2}$ Now substitute this value of c into the second equation to find k: $k = \left(-\frac{3}{2}\right)(4)$ $k = -\frac{12}{2}$ $k = -6$ Therefore, the value of k is -6. A common algebraic trap is to mistakenly set the components equal with.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Algebraic Definition and Properties

Find the dot product $\mathbf{u} \cdot \mathbf{v}$ for $\mathbf{u} = \langle 4, -3 \rangle$ and $\mathbf{v} = \langle 2, 5 \rangle$.

a. -7 ✓

b. 23

c. $\langle 8, -15 \rangle$

d. 1

$\mathbf{u} \cdot \mathbf{v} = u_1 v_1 + u_2 v_2 = (4)(2) + (-3)(5) = 8 - 15 = -7$.

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22 [Foundational] • Target: Algebraic Definition and Properties

What is the value of $\mathbf{v} \cdot \mathbf{v}$ for any vector $\mathbf{v}$?

a. $||\mathbf{v}||^2$ ✓

b. $||\mathbf{v}||$

c. 0

d. 1

$\mathbf{v} \cdot \mathbf{v} = v_1^2 + v_2^2 = ||\mathbf{v}||^2$.

23 [Foundational] • Target: Algebraic Definition and Properties

Which property does NOT hold for the dot product?

a. $\mathbf{u} \cdot \mathbf{v} = \mathbf{v} \cdot \mathbf{u}$ (Commutative)

b. $c(\mathbf{u} \cdot \mathbf{v}) = (c\mathbf{u}) \cdot \mathbf{v}$

c. $\mathbf{u} \cdot (\mathbf{v} + \mathbf{w}) = \mathbf{u} \cdot \mathbf{v} + \mathbf{u} \cdot \mathbf{w}$

d. $\mathbf{u} \cdot (\mathbf{v} \cdot \mathbf{w}) = (\mathbf{u} \cdot \mathbf{v}) \cdot \mathbf{w}$ **(Associative)** ✓

The dot product of two vectors is a scalar, so $\mathbf{v} \cdot \mathbf{w}$ is a scalar. You cannot dot a vector with a scalar, so associativity of dot products is undefined.

24 [Foundational] • Target: Geometric Definition and Angle Between Vectors

State the formula for the cosine of the angle θ between two non-zero vectors $\mathbf{u}$ and $\mathbf{v}$.

a. $\cos\theta = \frac{\mathbf{u} \cdot \mathbf{v}}{||\mathbf{u}||||\mathbf{v}||}$ ✓

b. $\cos\theta = \frac{||\mathbf{u}||||\mathbf{v}||}{\mathbf{u} \cdot \mathbf{v}}$

c. $\cos\theta = \mathbf{u} \cdot \mathbf{v}$

d. $\cos\theta = \frac{\mathbf{u} + \mathbf{v}}{||\mathbf{u} + \mathbf{v}||}$

From $\mathbf{u} \cdot \mathbf{v} = ||\mathbf{u}||||\mathbf{v}||\cos\theta$, we have $\cos\theta = \frac{\mathbf{u} \cdot \mathbf{v}}{||\mathbf{u}||||\mathbf{v}||}$.

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25 [Foundational] • Target: Geometric Definition and Angle Between Vectors

Find the angle between $\mathbf{u} = \langle 1, 1 \rangle$ and $\mathbf{v} = \langle 0, 2 \rangle$.

a. 45° ✓

b. 90°

c. 30°

d. 60°

$\mathbf{u} \cdot \mathbf{v} = 0 + 2 = 2$. $||\mathbf{u}|| = \sqrt{2}$, $||\mathbf{v}|| = 2$. $\cos\theta = \frac{2}{2\sqrt{2}} = \frac{1}{\sqrt{2}} \Rightarrow \theta = 45^\circ$.

26 [Foundational] • Target: Geometric Definition and Angle Between Vectors

If the dot product $\mathbf{u} \cdot \mathbf{v} < 0$, what type of angle is θ?

a. **Obtuse ($90^\circ < \theta \leq 180^\circ$)** ✓

b. Acute ($0^\circ \leq \theta < 90^\circ$)

c. Right (90°)

d. Straight (180°)

Since $\cos\theta = \frac{\mathbf{u} \cdot \mathbf{v}}{||\mathbf{u}||||\mathbf{v}||}$, a negative dot product implies $\cos\theta < 0$, which corresponds to an obtuse angle.

27 [Foundational] • Target: Orthogonality and Parallel Vectors

Two non-zero vectors $\mathbf{u}$ and $\mathbf{v}$ are orthogonal (perpendicular) if and only if:

a. $\mathbf{u} \cdot \mathbf{v} = 0$ ✓

b. $\mathbf{u} \cdot \mathbf{v} = 1$

c. $\mathbf{u} \times \mathbf{v} = \mathbf{0}$

d. $||\mathbf{u}|| = ||\mathbf{v}||$

$\mathbf{u} \cdot \mathbf{v} = ||\mathbf{u}||||\mathbf{v}||\cos(90^\circ) = 0$.

28 [Foundational] • Target: Orthogonality and Parallel Vectors

Find k such that $\mathbf{u} = \langle 3, k \rangle$ is orthogonal to $\mathbf{v} = \langle 4, 6 \rangle$.

a. -2 ✓

b. 2

c. $-1/2$

d. 12

$\mathbf{u} \cdot \mathbf{v} = 3(4) + 6k = 12 + 6k = 0 \Rightarrow 6k = -12 \Rightarrow k = -2$.

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Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 **[Intermediate]** • **Target: Orthogonality and Parallel Vectors**

Which pair of vectors is parallel?

a. $\langle 2, -3 \rangle$ **and** $\langle -6, 9 \rangle$ ✓

b. $\langle 2, 3 \rangle$ and $\langle 3, 2 \rangle$

c. $\langle 1, 0 \rangle$ and $\langle 0, 1 \rangle$

d. $\langle 4, 1 \rangle$ and $\langle -1, 4 \rangle$

$\langle -6, 9 \rangle = -3\langle 2, -3 \rangle$. A vector is parallel to another if it is a scalar multiple of it.

30 **[Intermediate]** • **Target: Vector Projections and Work**

State the formula for the vector projection of $\mathbf{u}$ onto $\mathbf{v}$ ($\text{proj}_{\mathbf{v}}\mathbf{u}$).

a. $\left(\frac{\mathbf{u} \cdot \mathbf{v}}{||\mathbf{v}||^2}\right)\mathbf{v}$ ✓

b. $\left(\frac{\mathbf{u} \cdot \mathbf{v}}{||\mathbf{u}||^2}\right)\mathbf{u}$

c. $\frac{\mathbf{u} \cdot \mathbf{v}}{||\mathbf{v}||}$

d. $(\mathbf{u} \cdot \mathbf{v})\mathbf{v}$

The vector projection is the scalar component $\frac{\mathbf{u} \cdot \mathbf{v}}{||\mathbf{v}||}$ multiplied by the unit vector $\frac{\mathbf{v}}{||\mathbf{v}||}$, yielding $\left(\frac{\mathbf{u} \cdot \mathbf{v}}{||\mathbf{v}||^2}\right)\mathbf{v}$.

31 **[Intermediate]** • **Target: Vector Projections and Work**

Find $\text{proj}_{\mathbf{v}}\mathbf{u}$ for $\mathbf{u} = \langle 6, 2 \rangle$ and $\mathbf{v} = \langle 4, 0 \rangle$.

a. $\langle 6, 0 \rangle$ ✓

b. $\langle 4, 0 \rangle$

c. $\langle 0, 2 \rangle$

d. $\langle 24, 0 \rangle$

$\mathbf{u} \cdot \mathbf{v} = 24$. $||\mathbf{v}||^2 = 16$. $\text{proj}_{\mathbf{v}}\mathbf{u} = \frac{24}{16}\langle 4, 0 \rangle = \frac{3}{2}\langle 4, 0 \rangle = \langle 6, 0 \rangle$.

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32 [Intermediate] • Target: Vector Projections and Work

A force $\mathbf{F} = \langle 30, 40 \rangle$ N moves a cart along the displacement $\mathbf{d} = \langle 5, 0 \rangle$ m. Calculate the work done.

a. 150 J ✓

b. 200 J

c. 350 J

d. 250 J

$W = \mathbf{F} \cdot \mathbf{d} = (30)(5) + (40)(0) = 150$ J.

33 [Intermediate] • Target: Orthogonality and Parallel Vectors

Find the vector projection of $\mathbf{u} = \langle 3, 4 \rangle$ onto $\mathbf{v} = \langle 2, 0 \rangle$.

a. $\langle 0, 3 \rangle$

b. $\langle 4, 0 \rangle$

c. $\langle 3, 0 \rangle$ ✓

d. $\langle 3, 4 \rangle$

$\text{proj}_{\mathbf{v}}\mathbf{u} = \frac{\mathbf{u} \cdot \mathbf{v}}{\mathbf{v} \cdot \mathbf{v}}\mathbf{v} = \frac{3}{2}\langle 2, 0 \rangle = \langle 3, 0 \rangle$.

34 [Intermediate] • Target: Vector Projections and Work

Find the vector projection of $\mathbf{u} = \langle 4, 3 \rangle$ onto $\mathbf{v} = \langle 0, 2 \rangle$.

a. $\langle 0, 3 \rangle$ ✓

b. $\langle 3, 0 \rangle$

c. $\langle 1, 3 \rangle$

d. $\langle 4, 3 \rangle$

$\text{proj}_{\mathbf{v}}\mathbf{u} = \frac{\mathbf{u} \cdot \mathbf{v}}{\mathbf{v} \cdot \mathbf{v}}\mathbf{v} = \frac{3}{2}\langle 0, 2 \rangle = \langle 0, 3 \rangle$.

35 [Intermediate] • Target: Computing dot products and angles

Find the vector projection of $\mathbf{u} = \langle 8, 6 \rangle$ onto $\mathbf{v} = \langle 0, 4 \rangle$.

a. $\langle 6, 0 \rangle$

b. $\langle 0, 6 \rangle$ ✓

c. $\langle 8, 6 \rangle$

d. $\langle 1, 6 \rangle$

$\text{proj}_{\mathbf{v}}\mathbf{u} = \frac{\mathbf{u} \cdot \mathbf{v}}{\mathbf{v} \cdot \mathbf{v}}\mathbf{v} = \frac{3}{2}\langle 0, 4 \rangle = \langle 0, 6 \rangle$.

36 [Intermediate] • Target: Projections and work

Find the vector projection of $\mathbf{u} = \langle 15, 8 \rangle$ onto $\mathbf{v} = \langle 0, 3 \rangle$.

a. $\langle 0, 8 \rangle$ ✓

b. $\langle 1, 8 \rangle$

c. $\langle 8, 0 \rangle$

d. $\langle 15, 8 \rangle$

$\text{proj}_{\mathbf{v}}\mathbf{u} = \frac{\mathbf{u} \cdot \mathbf{v}}{\mathbf{v} \cdot \mathbf{v}}\mathbf{v} = \frac{8}{3}\langle 0, 3 \rangle = \langle 0, 8 \rangle$.

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37 [Intermediate] • Target: Properties of the dot product

Find the vector projection of $\mathbf{u} = \langle 24, 7 \rangle$ onto $\mathbf{v} = \langle 0, 4 \rangle$.

a. $\langle 1, 7 \rangle$

b. $\langle 0, 7 \rangle$ ✓

c. $\langle 24, 7 \rangle$

d. $\langle 7, 0 \rangle$

$\text{proj}_{\mathbf{v}}\mathbf{u} = \frac{\mathbf{u} \cdot \mathbf{v}}{\mathbf{v} \cdot \mathbf{v}}\mathbf{v} = \frac{7}{4}\langle 0, 4 \rangle = \langle 0, 7 \rangle$.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Angle classification • Cross-Lesson: 2.02 Radian Measure and Arc Length

Find the vector projection of $\mathbf{u} = \langle 20, 21 \rangle$ onto $\mathbf{v} = \langle 4, 0 \rangle$.

a. $\langle 20, 0 \rangle$ ✓

b. $\langle 21, 0 \rangle$

c. $\langle 0, 20 \rangle$

d. $\langle 20, 21 \rangle$

$\text{proj}_{\mathbf{v}}\mathbf{u} = \frac{\mathbf{u} \cdot \mathbf{v}}{\mathbf{v} \cdot \mathbf{v}}\mathbf{v} = 5\langle 4, 0 \rangle = \langle 20, 0 \rangle$.

39 [Challenging] • Target: Scalar projection and further applications • Cross-Lesson: 2.03 Graphs of Sine and Cosine Functions

Find the vector projection of $\mathbf{u} = \langle 21, 20 \rangle$ onto $\mathbf{v} = \langle 0, 4 \rangle$.

a. $\langle 21, 20 \rangle$

b. $\langle 20, 0 \rangle$

c. $\langle 1, 20 \rangle$

d. $\langle 0, 20 \rangle$ ✓

$\text{proj}_{\mathbf{v}}\mathbf{u} = \frac{\mathbf{u} \cdot \mathbf{v}}{\mathbf{v} \cdot \mathbf{v}}\mathbf{v} = 5\langle 0, 4 \rangle = \langle 0, 20 \rangle$.

40 [Challenging] • Target: Decomposing a vector into parallel and perpendicular parts • Cross-Lesson: 2.04 Law of Sines and Law of Cosines

Find the vector projection of $\mathbf{u} = \langle 6, 8 \rangle$ onto $\mathbf{v} = \langle 3, 0 \rangle$.

a. $\langle 7, 0 \rangle$

b. $\langle 6, 0 \rangle$ ✓

c. $\langle 0, 6 \rangle$

d. $\langle 6, 8 \rangle$

$\text{proj}_{\mathbf{v}}\mathbf{u} = \frac{\mathbf{u} \cdot \mathbf{v}}{\mathbf{v} \cdot \mathbf{v}}\mathbf{v} = 2\langle 3, 0 \rangle = \langle 6, 0 \rangle$.

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41 [Challenging] • Target: Deriving the dot product angle formula • Cross-Lesson: 3.01 The Unit Circle and Exact Values

Find the vector projection of $\mathbf{u} = \langle 5, 12 \rangle$ onto $\mathbf{v} = \langle 1, 0 \rangle$.

a. $\langle 0, 5 \rangle$

b. $\langle 5, 0 \rangle$ ✓

c. $\langle 6, 0 \rangle$

d. $\langle 5, 12 \rangle$

$\text{proj}_{\mathbf{v}}\mathbf{u} = \frac{\mathbf{u} \cdot \mathbf{v}}{\mathbf{v} \cdot \mathbf{v}}\mathbf{v} = 5\langle 1, 0 \rangle = \langle 5, 0 \rangle$.

42 [Challenging] • Target: The Cauchy-Schwarz inequality (preview) • Cross-Lesson: 3.02 Fundamental Trigonometric Identities

Find the vector projection of $\mathbf{u} = \langle 12, 5 \rangle$ onto $\mathbf{v} = \langle 0, 1 \rangle$.

a. $\langle 1, 5 \rangle$

b. $\langle 0, 5 \rangle$ ✓

c. $\langle 5, 0 \rangle$

d. $\langle 12, 5 \rangle$

$\text{proj}_{\mathbf{v}}\mathbf{u} = \frac{\mathbf{u} \cdot \mathbf{v}}{\mathbf{v} \cdot \mathbf{v}}\mathbf{v} = 5\langle 0, 1 \rangle = \langle 0, 5 \rangle$.

43 [Challenging] • Target: Dot products with variable components • Cross-Lesson: 3.03 Sum and Difference Formulas

Find the vector projection of $\mathbf{u} = \langle 9, 12 \rangle$ onto $\mathbf{v} = \langle 3, 0 \rangle$.

a. $\langle 0, 9 \rangle$

b. $\langle 9, 12 \rangle$

c. $\langle 9, 0 \rangle$ ✓

d. $\langle 10, 0 \rangle$

$\text{proj}_{\mathbf{v}}\mathbf{u} = \frac{\mathbf{u} \cdot \mathbf{v}}{\mathbf{v} \cdot \mathbf{v}}\mathbf{v} = 3\langle 3, 0 \rangle = \langle 9, 0 \rangle$.

44 [Challenging] • Target: A capstone dot product application • Cross-Lesson: 3.04 Double-Angle Formulas and Trigonometric Equations

Find the vector projection of $\mathbf{u} = \langle 12, 9 \rangle$ onto $\mathbf{v} = \langle 0, 3 \rangle$.

a. $\langle 9, 0 \rangle$

b. $\langle 0, 9 \rangle$ ✓

c. $\langle 12, 9 \rangle$

d. $\langle 1, 9 \rangle$

$\text{proj}_{\mathbf{v}}\mathbf{u} = \frac{\mathbf{u} \cdot \mathbf{v}}{\mathbf{v} \cdot \mathbf{v}}\mathbf{v} = 3\langle 0, 3 \rangle = \langle 0, 9 \rangle$.

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45 [Challenging] • Target: Algebraic Definition and Properties • Cross-Lesson: 4.01 Circles and Parabolas

Find the vector projection of $\mathbf{u} = \langle 8, 15 \rangle$ onto $\mathbf{v} = \langle 2, 0 \rangle$.

a. $\langle 8, 15 \rangle$

b. $\langle 9, 0 \rangle$

c. $\langle 8, 0 \rangle$ ✓

d. $\langle 0, 8 \rangle$

$\text{proj}_{\mathbf{v}}\mathbf{u} = \frac{\mathbf{u} \cdot \mathbf{v}}{\mathbf{v} \cdot \mathbf{v}}\mathbf{v} = 4\langle 2, 0 \rangle = \langle 8, 0 \rangle$.

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